Force and Acceleration

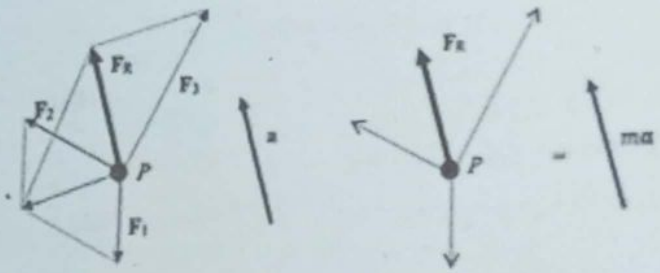
- **Newton's Laws of Motion**

  Newton's second law of motion may be written in mathematical form as

  $$F = ma$$

  This equation is referred to as the equation of motion.

- Equations of Motion of a Particle



(a) Free-body diagram (b) Kinetic diagram

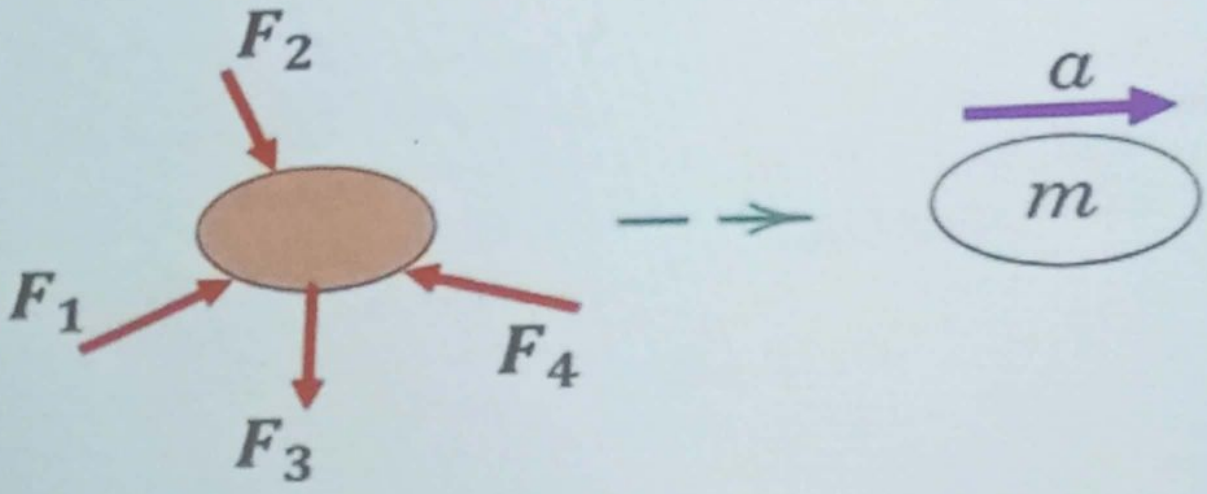
Consider the particle P, shown in Figure l, which has a,mass m and .is,subjected

Force and Acceleration

Equation of Motion

For a system of Forces

A particle of mass m is subjected to the action of concurrent forces F_1, F_2, F_3,... whose vector sum is ΣF



$$\Sigma F = ma$$

Force and Acceleration

Two Types of Dynamics Problems

Type I

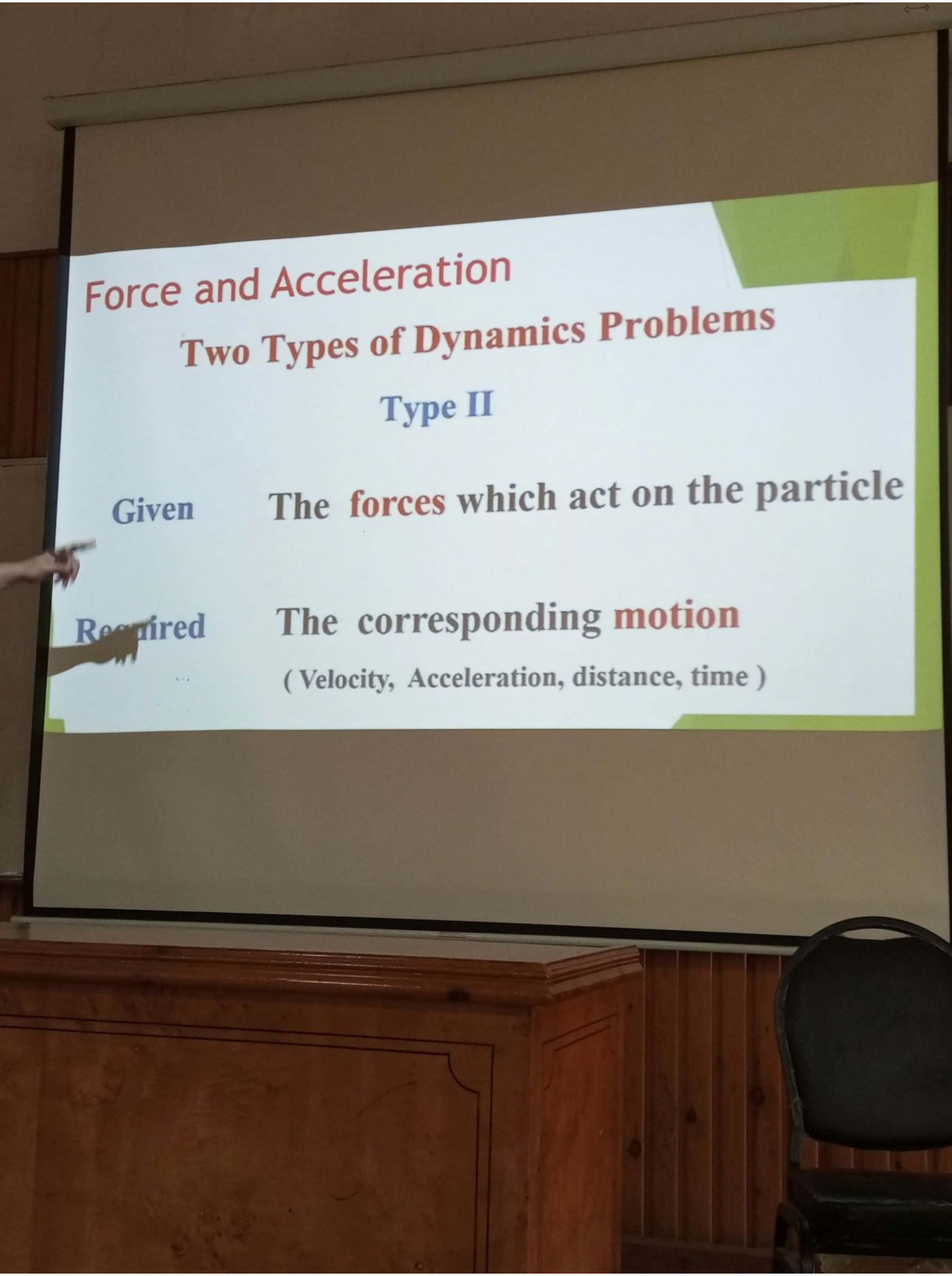
Given **The acceleration:**

is either known or can be determined from kinematic conditions.

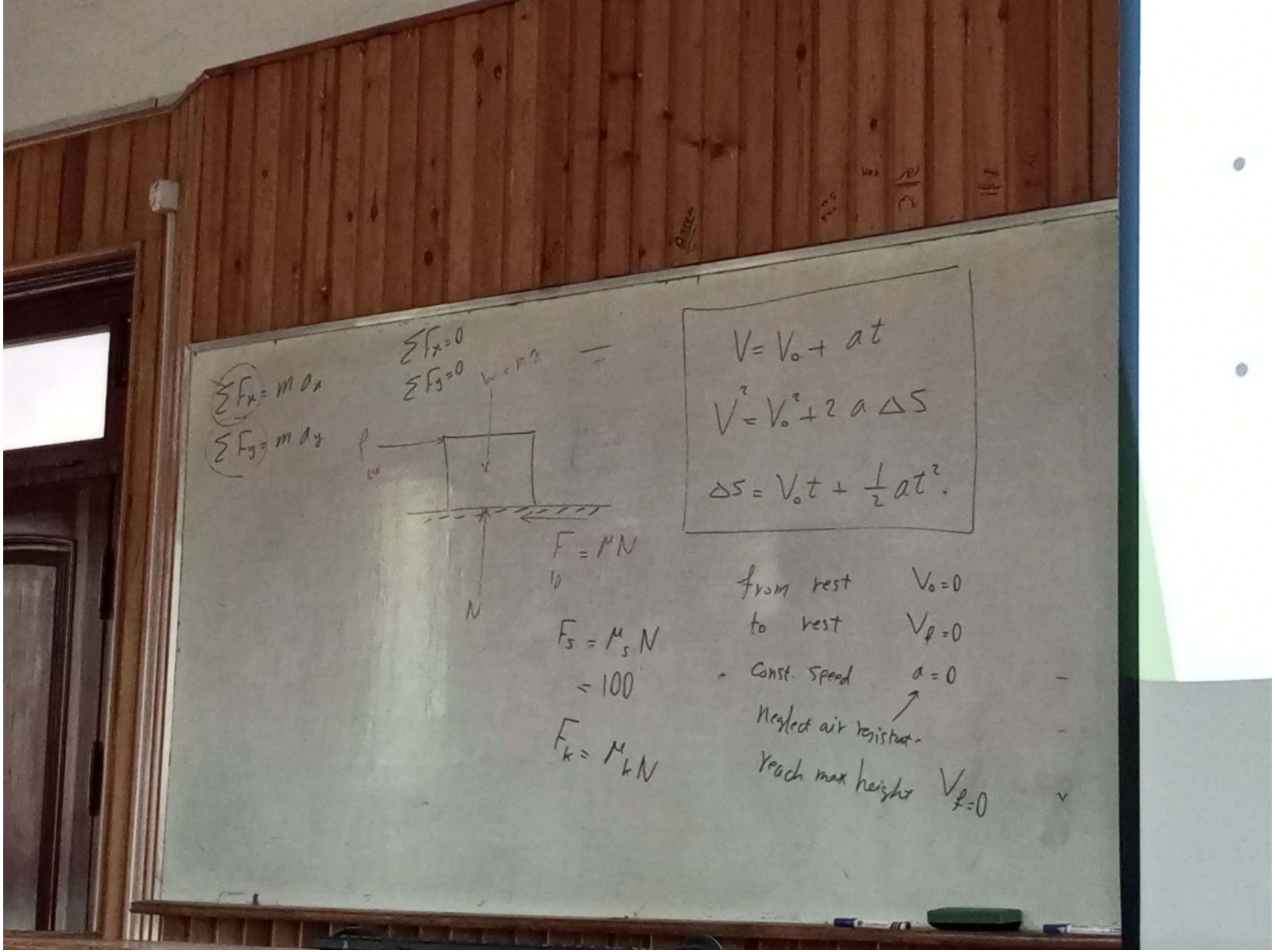
Required **The force which act on the particle:**

is determined from the Equation of Motion

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Force and Acceleration
Two Types of Dynamics Problems
Type II
Given
The forces which act on the particle
Required
The corresponding motion
(Velocity, Acceleration, distance, time)

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$\sum F_x = ma_x$
$\sum F_y = ma_y$
$\sum F_x = 0$
$\sum F_y = 0$
$W = mg$
F_{ap}
$F = \mu N$
N
$F_s = \mu_s N$
$= 100$
$F_k = \mu_k N$
$V = V_0 + at$
$V^2 = V_0^2 + 2a\Delta s$
$\Delta s = V_0 t + \frac{1}{2}at^2$.
from rest $V_0 = 0$
to rest $V_f = 0$
Const. speed $a = 0$
Neglect air resistance
reach max height $V_f = 0$

Force and Acceleration

Rectilinear Motion

If the motion of a mass m is restricted to the *x-direction*, then

$$a_y = a_z = 0$$

$$\sum \boldsymbol{F} = \boldsymbol{ma} \rightarrow \quad \sum F_x = ma_x$$

$$\sum F_y = 0$$

$$\sum F_z = 0$$

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Force and Acceleration

Curvilinear Motion

Rectangular Coordinates

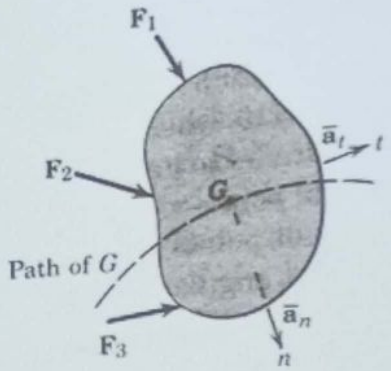
$$\sum F_x = ma_x$$
$$\sum F_y = ma_y$$

Normal and Tangential Coordinates

$$\sum F_n = ma_n$$
$$\sum F_t = ma_t$$

Polar Coordinates

$$\sum F_r = ma_r$$
$$\sum F_\theta = ma_\theta$$



Free-Body Diagram

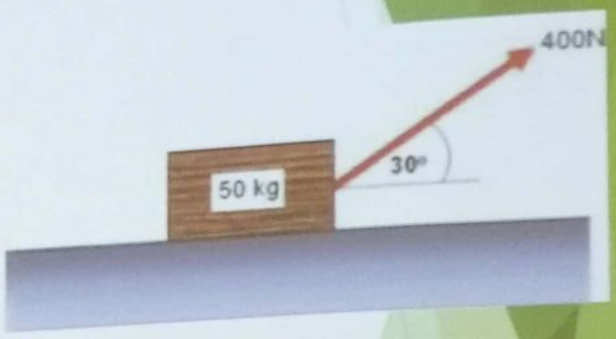
$$v_r = \dot{r}$$
$$v_\theta = r\dot{\theta}$$

$$a_r = \ddot{r} - r\dot{\theta}^2$$
$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$$

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Force and Acceleration

Ex:- The 50-kg crate shown in Figure rests on a horizontal plate for which the coefficient of kinetic friction is $\mu_k = 0.3$. If the crate is subjected to a 400-N towing force as shown, determine the velocity of the crate in 5 seconds starting from rest.

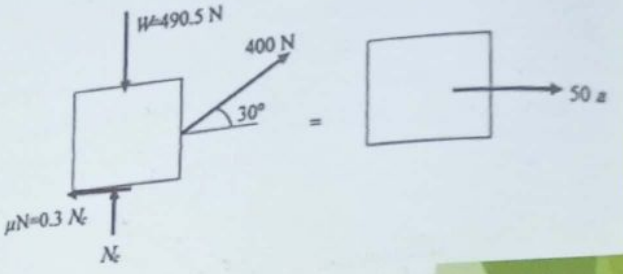


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Force and Acceleration

Step 1: The weight of the crate is

$$W = mg = 50\ \text{kg}(9.81\ \text{m/s}^2) = 490.5\ \text{N}$$



Step 2: Applying the equations of motion

$$\sum F_x = ma_x; \qquad 400\cos 30^\circ - 0.3\,N_c = 50\,a$$

$$\sum F_y = ma_y; \qquad N_c - 490.5 + 400\sin 30^\circ = 0$$

Solving the two equations

$$N_c = 290.5\ \text{N}$$

$$a = 5.19\ \text{m/s}^2$$

Step 3: Use the Equations of kinematics.

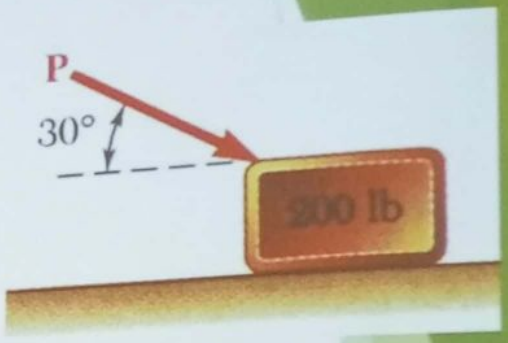
$$v = v_0 + at$$

$$= 0 + 5.19\,(5) = 26.0\ \text{m/s}.$$

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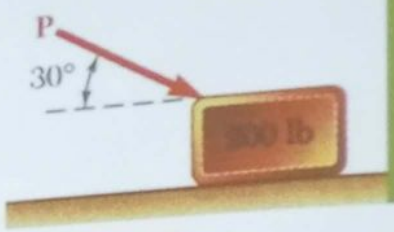
Force and Acceleration

Example:- A 200-lb block rests on a horizontal plane. Find the magnitude of the force *P* required to give the block an acceleration or 10 ft/s^2 to the right. The coefficient of kinetic friction between the block and plane is $\mu_k = 0.25$.



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Force and Acceleration



SOLUTION:

- Resolve the equation of motion for the block into two rectangular component equations.

$\sum F_x = ma$:

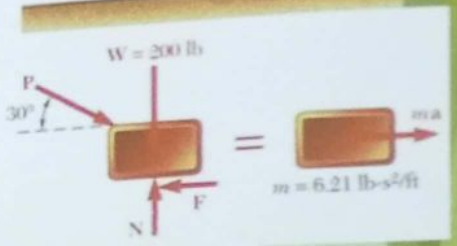
$$P\cos 30° - 0.25N = \left(6.21\,\text{lb}\cdot\text{s}^2/\text{ft}\right)\left(10\,\text{ft}/\text{s}^2\right)$$
$$= 62.1\,\text{lb}$$

$\sum F_y = 0$:

$$N - P\sin 30° - 200\,\text{lb} = 0$$

- Unknowns consist of the applied force P and the normal reaction N from the plane. The two equations may be solved for these unknowns.

$$N = P\sin 30° + 200\,\text{lb}$$
$$P\cos 30° - 0.25(P\sin 30° + 200\,\text{lb}) = 62.1\,\text{lb}$$
$$\boxed{P = 151\,\text{lb}}$$

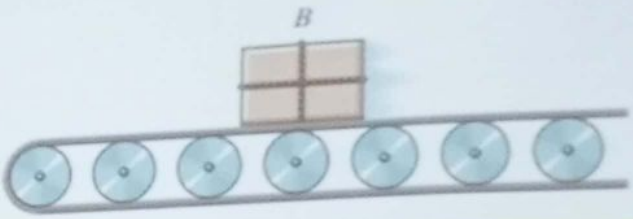


$$m = \frac{W}{g} = \frac{200\,\text{lb}}{32.2\,\text{ft}/\text{s}^2}$$
$$= 6.21\frac{\text{lb}\cdot\text{s}^2}{\text{ft}}$$

$$F = \mu_k N$$
$$= 0.25N$$

Force and Acceleration

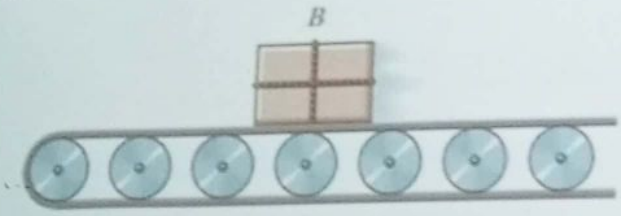
Ex:- The conveyor belt is moving at 4 m/s. If the coefficient of static friction between the conveyor and the 10-kg package B is μs = 0.2, determines the shortest time the belt can stop so that the package does not slide on the belt.



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Force and Acceleration

Ex:- The conveyor belt is moving at 4 m/s. If the coefficient of static friction between the conveyor and the 10-kg package B is μs = 0.2, determines the shortest time the belt can stop so that the package does not slide on the belt.



Given: $m = 10\ kg$

$v_0 = 4\ m/s$

$v_f = 0$

$\mu_s = 0.2$

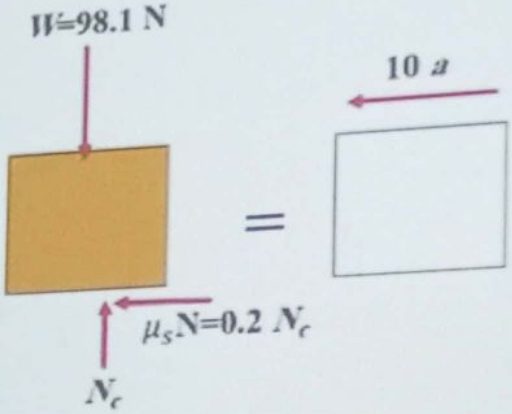
Required: $t = ?$

Missing: a

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Force and Acceleration

$W = mg = 10\ kg(9.81\ m/s^2) = 98.1\ N$



The frictional force has a magnitude of $F = \mu_s N$

The equations of motion

$$\sum F_y = ma_y\ ;\rightarrow \qquad N_c - 98.1 = 0 \quad \rightarrow \therefore\ N_c = 98.1\ N$$

$$\sum F_x = ma_x\ ;\rightarrow \qquad -0.2\,N_c = -10a \quad \rightarrow \therefore\ a = 1.962\ m/s^2$$

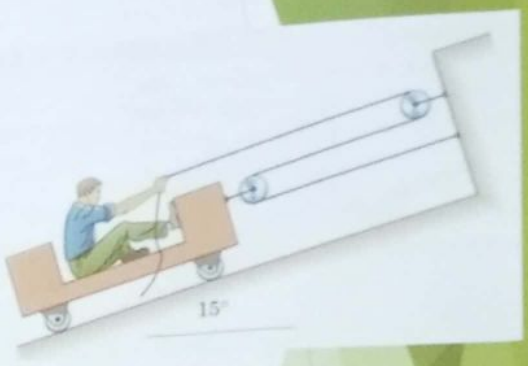
Kinematics:

$$v_f = v_0 + at\ ;\rightarrow \qquad 0 = 4 - 1.962t \quad \rightarrow \quad t = 2.04\ s$$

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Force and Acceleration

Ex:- A man pulls himself up the 15° incline by the method shown. if the combined mass of the man and cart is 100kg, determine the acceleration of cart cart if the man exerts a pull of 250 N on the rope. neglect all friction and the mass of the rope, wheels and pulleys.



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Force and Acceleration

Solution

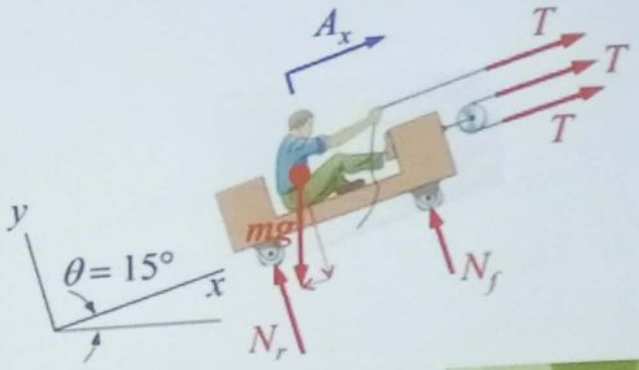
$$\Sigma F_x = mA_x$$

$$3T - mg\sin\theta = mA_x$$

$$\text{i.e., } A_x = \frac{3T - mg\sin\theta}{m}$$

$$= \frac{3\times 250 - 100\times 9.81\times \sin 15^\circ}{100}$$

$$= 4.96 \text{ m/s}^2$$



Force and Acceleration

Ex:- The van is traveling at 20 km/h when the coupling of the trailer at A fails. If the trailer has a mass of 250 kg and coasts 45 m before coming to rest, determine the constant horizontal force F created by rolling friction which causes the trailer to stop.



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Force and Acceleration

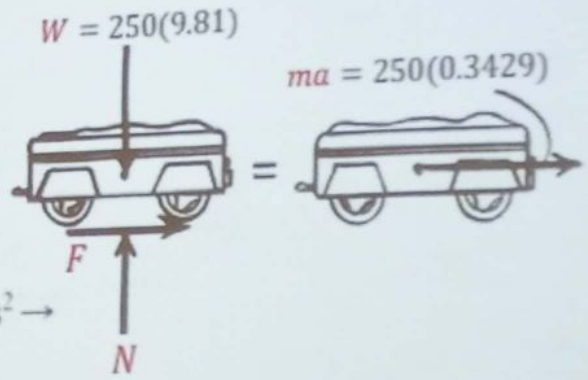
$$20\ \text{km/h} = \frac{20(10^3)}{3600} = 5.556\ \text{m/s}$$

$$\left(\pm\right) \quad v^2 = v_0^2 + 2a_c(s - s_0)$$

$$0 = 5.556^2 + 2(a)(45 - 0)$$

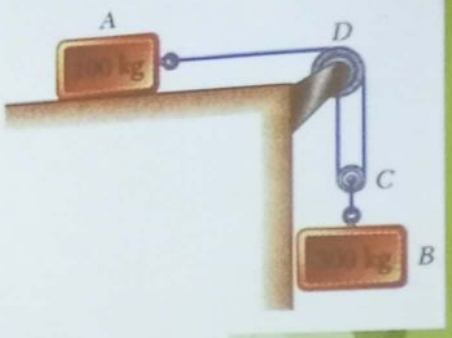
$$a = -0.3429\ \text{m/s}^2 = 0.3429\ \text{m/s}^2 \rightarrow$$

$$\pm \Sigma F_x = ma_x; \rightarrow \quad F = 250(0.3429) = 85.7\ \text{N}$$



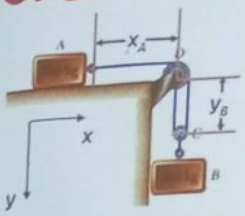
Force and Acceleration

Ex:- The two blocks shown start from rest. The horizontal plane and the pulley are frictionless, and the pulley is assumed to be of negligible mass. Determine the acceleration of each block and the tension in the cord.



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Force and Acceleration



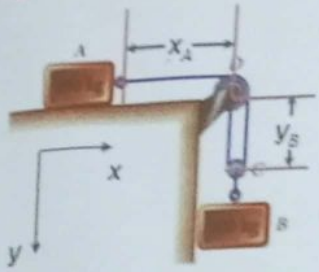
SOLUTION:

- Write the kinematic relationships for the Relative motions and accelerations of the blocks.

$$y_B = \frac{1}{2} x_A \qquad a_B = \frac{1}{2} a_A$$

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Force and Acceleration

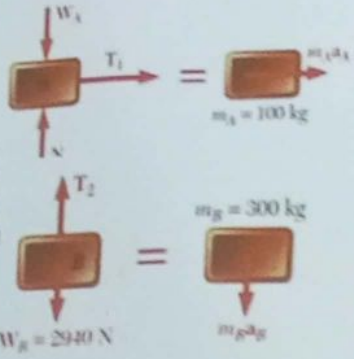


SOLUTION:

- Write the kinematic relationships for the Relative motions and accelerations of the blocks.

$$y_B = \frac{1}{2}x_A \qquad a_B = \frac{1}{2}a_A$$

- Write equations of motion for blocks and pulley.



$$\sum F_x = m_A a_A:$$

$$T_1 = (100\,\text{kg})a_A$$

$$\sum F_y = m_B a_B:$$

$$m_B g - T_2 = m_B a_B$$

$$(300\,\text{kg})(9.81\,\text{m/s}^2) - T_2 = (300\,\text{kg})a_B$$

$$T_2 = 2940\text{N} - (300\,\text{kg})a_B$$

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Force and Acceleration

- Combine kinematic relationships with equations of motion to solve for accelerations and cord tension.

$$y_B = \tfrac{1}{2}x_A \qquad a_B = \tfrac{1}{2}a_A$$

$$T_1 = (100\,\text{kg})a_A$$

$$T_2 = 2940\text{N} - (300\,\text{kg})a_B$$

$$= 2940\text{N} - (300\,\text{kg})\left(\tfrac{1}{2}a_A\right)$$

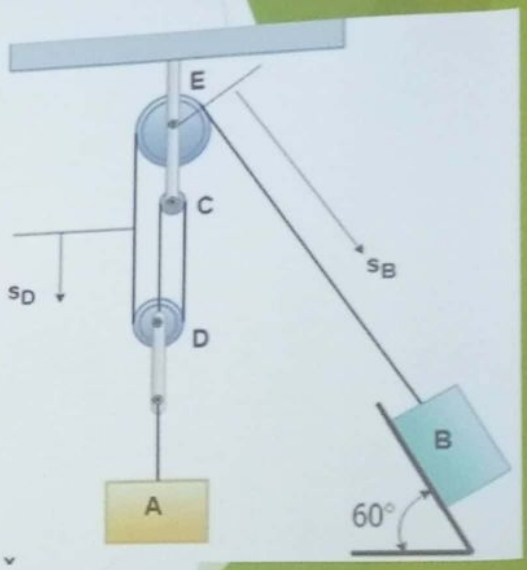
$$T_2 - 2T_1 = 0$$

$$2940\text{N} - (150\,\text{kg})a_A - 2(100\,\text{kg})a_A = 0$$

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Force and Acceleration

Ex:- Determine the required mass of block A so that when it is released from rest it moves the block B a distance S_B=0.75m up the inclined plane shown in the figure in time t=2 s. Neglect the mass of the pulleys and cords. Block B has a mass of m =5 kg and the coefficient of friction between the block and the inclined is 0.25.



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Force and Acceleration

The initial velocity V_0=0. The block moves a distance so=0.75 m in time t=2.0 s.

$$s_B = v_0 t + \frac{1}{2} a_B t^2$$

$$0.75 = 0 + \frac{1}{2} a_B (2)^2$$

$$a_B = 0.375 \ \ m/s^2.$$

